

Cody Kochmann

Principal Infrastructure Engineer, Security Developer & DevSecOps Enthusiast

443-538-9639 | kochmanncody@gmail.com | <https://github.com/CodyKochmann>

Summary

I'm a principal infrastructure engineer, security developer, and DevSecOps enthusiast. Whether you need to accelerate your developer workflows through automation, harden your infrastructure against state-sponsored attacks, or design high-performance distributed systems, I have a proven track record of driving mission-critical company goals.

Technical Skills

- **Infrastructure & Cloud:** Kubernetes (EKS, AKS, GKE), Terraform, OpenTofu, AWS, Azure, Linux, CloudFormation, ArgoCD, Docker
- **Security & Operations:** DevSecOps, Istio, Falco, Vault, Trivy, Threat Detection, Vulnerability Management, RBAC, Grafana, ELK/HELK Stack
- **Languages & Automation:** Python, Golang, Rust, Bash, CI/CD (GitLab, GitHub Actions, Buildkite, Argo), Infrastructure as Code (IaC)
- **Agentic AI & LLMs:** Claude, LangChain, LangGraph, Ollama, Mistral, Qwen, Hermes, OpenAI, Mem0, Antigravity, OpenClaw, Oh-My-Pi, RAG (Retrieval-Augmented Generation), Vector Databases, Prompt Engineering, Agentic Frameworks, LLM Sandboxing & Harnesses, LLMops, MLOps, AI Security, LLM Guardrails, On-Premise Inferencing, AIOps

Experience

NICO - 2025 to Present

Roles: Principal Infrastructure Engineer

Programming Languages: Bash, Python, YAML

Technologies: Azure, Kubernetes (AKS), Terraform, OpenTofu, ArgoCD, Argo Workflows, Grafana Stack, Istio, Falco, HashiCorp Vault (OpenBao), SealedSecrets, Zot, Trivy, Agentic AI (Claude, Hermes)

Achievements:

- **Security & Operations:** Hardened network traffic by deploying an Istio service mesh, configured Falco for real-time intrusion detection (IDS), and established strict RBAC across all environments.
 - **Supply Chain Security:** Secured the software supply chain by building isolated build environments, integrating Trivy for automated container vulnerability scanning, and managing the Zot registry. Safely managed credentials utilizing Vault and SealedSecrets.
 - **Cloud Infrastructure:** Led the comprehensive overhaul of the Third-Party Risk Management (TPRM) Azure infrastructure, establishing a mature, highly resilient posture rooted in industry best practices.
 - **Kubernetes & IaC:** Architected and executed the migration of all live infrastructure to Kubernetes, fully codifying the environment into modular, reusable Terraform components.
 - **AI-Accelerated Operations:** Leveraged Agentic AI tooling as a force multiplier to rapidly architect, test, and deploy infrastructure as code, significantly accelerating project delivery timelines.
 - **GitOps & Automation:** Championed the adoption of immutable infrastructure principles by implementing GitOps workflows with ArgoCD.
 - **Observability & Monitoring:** Designed and deployed centralized observability and logging pipelines using the complete Grafana stack.
-

Agentic AI Researcher & Consultant - 2024

Roles: Agentic AI Researcher (Collaborative Initiative with former NASA colleagues)

Programming Languages: Python, Bash

Technologies: Large Language Models (LLMs), Agentic Frameworks, Prompt Engineering, Vector Databases

Achievements:

- Partnered with former NASA engineering colleagues to conduct extensive research into Large Language Model (LLM) design, training methodologies, and practical utilization for software engineering in secure environments.
 - Designed and tested secure sandboxing methods to safely provide LLM capabilities within highly sensitive and restricted enterprise networks.
 - Analyzed the security risks and hallucination vectors of deploying LLMs in enterprise environments, developing strategies for safe, air-gapped adoption.
 - Collaborated on critical research initiatives focused on optimizing AI operational spend and reducing infrastructure overhead.
 - Researched and implemented local GPU optimization techniques to maximize the performance of on-premise inferencing.
 - Mastered advanced prompting techniques and agentic workflows (utilizing tools like Antigravity, Claude and Ollama) which directly translated into immense productivity multipliers for future infrastructure automation roles.
-

Strake - September 2023 to December 2023

Strake closed business in December 2023.

Roles: Principal Infrastructure Engineer

Programming Languages: Make, Python, Bash

Technologies: GitHub, CloudFormation, CodeDeploy, BuildKite, Postgres, SQLite3, SQS, Lambda

Achievements:

- Drove the development of the tools and workflows needed to generate ephemeral development environments for engineers to safely perform isolated experiments.
 - Designed and implemented tools that provided deep insights to every corner of the company's infrastructure, development patterns, and communication effectiveness.
-

QOMPLX - September 2019 to July 2023

QOMPLX closed business in July 2023.

Roles: Principal Cloud Platform Automation Engineer, Project Lead, Kubernetes SME, Solutions Architect, Security Architect, Security Developer, Mentor

Programming Languages: Python, Golang, Rust, Make, Bash, Jinja, YAML

Technologies: GitLab, CI/CD Pipelines, CI Runners, Kubernetes, Docker, Terraform, Linux Kernel, rsyslog, syslog-ng, Postgres, SQLite3, perf, ebp, LVM2, Loki, Grafana, Kafka, Elasticsearch, Logstash, Kibana, Prometheus, Aqua, Sysdig, Twistlock, CrowdStrike, Linux Kernel, Digital Ocean, Linode, AWS (EKS, ECS, EC2, Lambda, CloudTrail, SQS, S3, Lambda, Config, IAM), Istio, Argo, Debian, Amazon Linux, CoreOS, Ubuntu, Fedora, FreeBSD, CentOS, VMWare

Achievements:

- **AWS Cloud Security & IAM:** Acted as Lead Security Architect, driving the migration to full RBAC-based access control. Designed our multi-account AWS Organization modeled after NASA's multi-account

architecture, and enforced custom IAM least-privilege policies for all internal engineering teams and external contractor groups.

- **Data Protection & Secrets Management:** Eradicated hardcoded credentials by migrating all application secrets into AWS Secrets Manager and HashiCorp Vault. Enforced strict encryption-at-rest across all S3 buckets and EBS volumes, and both mandated and implemented TLS and mTLS encryption-in-transit for all internal and external network communications.
 - **Security & Threat Management:** Drove security and vulnerability scanning operations company-wide. Identified and prevented active security breaches and remediated thousands of legacy infrastructure vulnerabilities.
 - **Infrastructure & Kubernetes:** Spearheaded the company's comprehensive migration to Kubernetes as a Subject Matter Expert (SME) and Security Architect.
 - **Client Operations:** Salvaged major client relationships by directly resolving complex product issues in live production environments and managing the security of specialized on-prem deployments.
 - **Automation & CI/CD (GitLab):** Led the design and development of the company's GitLab organization and CI/CD pipelines. Embedded security scanning (SAST/SCA) into the deployment process and architected the RBAC controls for GitLab CI/CD pipelines to securely manage our entire multi-account AWS Organization.
 - **Observability:** Architected centralized log collection systems, providing unprecedented levels of observability across all operational layers of the company.
 - **Technical Leadership:** Acted as the platform team's Scrum Master and held weekly lectures for developers on defensive coding, infrastructure optimization, and high-performance architecture.
-

EMBEDDED FLIGHT SYSTEMS INC (NASA) - January 2017 to July 2019

Roles: Network Operations Engineer, Security Developer, Mentor

Programming Languages: Python, Golang, Rust, Bash

Technologies: Docker, Kubernetes, rocket, Twistlock, Sysdig, Aqua, HELK, Security Onion, OpenShift, Kubernetes, Docker, Amazon Lambda, Amazon ECS, Amazon EKS, Digital Ocean, Google GKE, Systemd, RPM, Debian, Amazon Linux, CoreOS, Ubuntu, Fedora, FreeBSD, CentOS, Cisco, VMWare

Achievements:

- **Hybrid Cloud Infrastructure (AWS & IaC):** Spearheaded the initiative to integrate NASA's on-premise networking with AWS. Engineered the cloud equivalents to NASA's stringent on-prem security standards, fully codifying the AWS infrastructure using Terraform.
 - **Security Architecture:** Designed and built a high-performance DDoS mitigation tool capable of protecting NASA's network from state-sponsored attacks.
 - **Vendor Risk Management:** Engineered automated security scanners that enabled third-party vendors to harden their own systems to meet the strict compliance requirements necessary to connect to our network.
 - **Container Architecture & Security:** Led the effort to migrate numerous legacy VM deployments to modern container-based solutions, and strictly enforced container vulnerability scanning across all organizations within the NASA AWS cloud environment.
 - **Cloud Security Engineering:** Authored and adopted various specialized cloud security tools to help NASA actively harden their AWS accounts against emerging threats.
 - **Black Box Security Analysis:** Performed deep security analysis on numerous black box solutions to evaluate their safety and implemented isolation strategies to keep them locked down when risks were identified.
 - **Security Operations & Observability:** Architected a high-performance distributed ELK cluster to ingest and analyze network logs across our hybrid cloud / on-prem network. Designed and implemented a multitude of netflow/packet capture/firewall log analysis tools to give the company deeper insight to AWS, on-prem, and container network traffic.
 - **Technical Leadership:** Taught developers about the principles of defensive coding and how to build highly resilient architectures that met NASA's reliability and security standards.
-

Open Source Development | June 2013 - Present

I am the lead author and architect for all of the following projects.

Project	Description
battle_tested	Fully automated Python fuzzer that highlights every crash a piece of code will raise in production.
generators	High-performance pipeline processing library written in pure Python.
graphdb	The fastest pure Python graph database available on PyPI.
queued	Turns functions into fully functional multi-core queued services.
strict_functions	Function decorators for more secure Python code execution.
stricttuple	Python tuples with Rust-style built-in rule enforcement.
litecollections	Python collection types except they're all backed by SQLite.

References

All of the following were solid people I had the honor to work with and would love to work with again one day.

Name	Title & Company	Contact
Marcin Pohl	Systems / Security Engineer @ NASA	marcin.pohl@nasa.gov
Adam Younce	CTO @ Strake	ayounce@ripcord.net
Beck Norris	Information Security / Compliance Expert @ Frontier Airlines	rrnorris33@gmail.com
Mason Walton	Automation Engineer @ QOMPLX	fin.mwalto7@gmail.com
Jason Hurley	Platform Team Lead @ QOMPLX	j.m.hurley@gmail.com
Nic Ladas	Senior Platform Architect @ QOMPLX	https://www.linkedin.com/in/nicladas/
Christopher Mishaga	CISO @ NASA	christopher.a.mishaga@nasa.gov